Manual

Overview

`dt-csp-fairness-tool` finds Individual Discriminatory Instances (IDIs) in pre-trained neural network models. An IDI is a pair of inputs identical except for a sensitive feature (e.g., gender, race, age) where the model's output differs by more than 0.05 — indicating the model treats people differently based on a protected characteristic.

Three methods are implemented and compared against a random-search baseline. The baseline samples real data rows and flips **all** sensitive features simultaneously (e.g. sex, race, and age together), consistent with the reference implementation in `lab4_solution.py`. All three methods use the same simultaneous flip definition so comparisons are like-for-like.

Method	Description
`dt`	**DT-only** — random seed sampling to label pairs, trains a DecisionTreeClassifier to identify high-discrimination feature regions, then samples more inputs from those regions
`csp`	**CSP-only** — uses CP-SAT (Google OR-Tools) to generate inputs across the feature domain with diversity constraints to avoid revisiting known discriminatory regions
`hybrid`	**Hybrid** — Phase 1 (DT seed sampling + rule extraction) followed by Phase 2 (CSP constrained to DT-identified regions)

Primary finding: DT is the strongest method, consistently and substantially beating the baseline. CSP and Hybrid are included as ablation comparisons and provide valuable evaluation content around the trade-offs of constraint-based vs data-driven search.

Setup

1. Clone the repository
2. Install dependencies: `pip install -r requirements.txt`
3. Ensure `dataset/` and `DNN/` folders contain the required files (see below)

> **Windows note:** Commands below use `python`. On some Windows setups the Python Launcher is invoked as `py` instead (e.g. `py main.py ...`). Use whichever works in your environment. On Linux/macOS `python3` may be required depending on your PATH.

Required Files

Datasets (`dataset/`):

Primary (required for main experiment):

- `processed_kdd.csv`
- `processed_adult.csv`
- `processed_compas.csv`
- `processed_german.csv`

- `processed_dutch.csv`
- `processed_credit_with_numerical.csv`

Ablation only (required for `--dataset all`):

- `processed_communities_crime.csv`
- `processed_law_school.csv`

Models (`DNN/`):

Primary:

- `model_processed_kdd_cleaned.h5`
- `model_processed_adult.h5`
- `model_processed_compas.h5`
- `model_processed_greman_cleaned.h5` ← note: typo in source filename, keep as-is
- `model_processed_dutch.h5`
- `model_processed_credit.h5`

Ablation only:

- `model_processed_communities_crime.h5`
- `model_processed_law_school_cleaned.h5`

Running the Tool

```
# Full experiment — all 8 datasets, all methods (recommended; expect several hours)
python main.py --dataset all --method all --budget 1000 --runs 20

# Full experiment saving to a separate folder (e.g. to preserve a previous run)
python main.py --dataset all --method all --budget 1000 --runs 20 --results-dir results_v2

# All three methods vs baseline on a single dataset (baseline shared, run once)
python main.py --dataset kdd --method all --budget 1000 --runs 20

# Multiple specific datasets (comma-separated, spaces optional)
python main.py --dataset kdd,adult,compas --method all --budget 1000 --runs 20

# Single method vs baseline
python main.py --dataset kdd --method dt     --budget 1000 --runs 20
python main.py --dataset kdd --method csp    --budget 1000 --runs 20
python main.py --dataset kdd --method hybrid --budget 1000 --runs 20

# Quick sanity check — 3 runs on one dataset to verify changes before a full run
python main.py --dataset adult --method all --runs 3 --results-dir results_test
```

> **Note:** `--dataset all` runs all 8 datasets (KDD, Adult, COMPAS, Dutch, Credit, German, Law School, Communities & Crime). Law School and Communities & Crime are ablation-only datasets; to run only

> the 6 primary datasets, either specify them individually or use a comma-separated list.

Arguments:

Argument	Default	Description
`--dataset`	required	Single dataset name, comma-separated list (e.g. `kdd,adult,compas`), or `all` (all 8 datasets)
`--method`	`hybrid`	`dt`, `csp`, `hybrid`, or `all`
`--budget`	`1000`	Total inputs generated per run
`--runs`	`20`	Independent trials per method (min 20 for meaningful Wilcoxon p-values)
`--seed-ratio`	`0.15`	Fraction of budget used for Phase 1 seed sampling (DT and Hybrid only)
`--results-dir`	`results`	Directory to write output CSVs — use a different path to avoid overwriting a previous run

Output

Results are saved incrementally — each run is written immediately so partial results survive an interrupted experiment. Output goes to `--results-dir` (default: `results/`):

- `<results-dir>/<dataset>_baseline.csv` — per-run IDI ratios for random search
- `<results-dir>/<dataset>_dt.csv` — per-run IDI ratios for DT-only
- `<results-dir>/<dataset>_csp.csv` — per-run IDI ratios for CSP-only
- `<results-dir>/<dataset>_hybrid.csv` — per-run IDI ratios for Hybrid
- `<results-dir>/summary.csv` — Wilcoxon comparison summary (means, std, p-values, timing); updated after each dataset completes when running `--dataset all`

IDI ratio = unique discriminatory inputs found / total inputs generated. Higher is better — a higher ratio means the method found more discrimination per unit of budget.

Configuration

Key parameters in `src/config.py`:

Parameter	Default	Description
`DEFAULT_BUDGET`	`1000`	Total inputs per run
`DEFAULT_SEED_RATIO`	`0.15`	Phase 1 seed fraction
`DT_MAX_DEPTH`	`4`	Decision tree max depth
`DIVERSITY_K`	`1`	Min features that must differ from any known discriminatory input (CSP diversity constraint)

Group Fairness Characterisation

A standalone script is provided to profile each model's group-level bias before running IDI search. It computes two metrics per sensitive feature:

- **Demographic Parity Difference (DPD)** — max difference in positive prediction rate across sensitive attribute groups. Does not require ground truth labels.
- **Equalized Odds Difference (EOD)** — max of TPR difference and FPR difference across groups. Requires the ground truth target column (already present in all dataset CSVs).

```
# Characterise all configured datasets
python characterise.py

# Single dataset
python characterise.py --dataset adult
```

Results are printed as a table per dataset and saved to `characterisation.csv` at the repo root. This script is intended as a one-time characterisation pass; re-running it produces the same output (no randomness involved).

Testing

The test suite covers all modules with 135 tests total. No real datasets or `.h5` model files are required — tests use `MockModel` fixtures and synthetic `pandas` DataFrames.

```
# Run all tests
python -m pytest tests/ -v

# Run per module
python -m pytest tests/test_dt.py       -v   # 33 tests
python -m pytest tests/test_csp.py      -v   # 47 tests
python -m pytest tests/test_hybrid.py   -v   # 24 tests
python -m pytest tests/test_evaluate.py -v   # 31 tests
```

scipy and numpy RuntimeWarnings (e.g. Wilcoxon on all-zero differences) are suppressed via `conftest.py` and do not indicate failures.
